

Table 1 stores the login accounts of every user of the system, including the System Administrator, Administrative Officer II, HRMO, and teaching and non-teaching personnel. It records the email address, the hashed password, the assigned role, and the account status, together with security details such as failed sign-in attempts, lockouts, and whether a password change or device verification is required. This ensures that only authorized persons can access the system.

Table 1.

Users Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
users	id (PK)	integer	No	Unique identifier of the record.
	email	string	No	Email address; unique regardless of letter case.
	password_hash	string	No	Bcrypt hash of the account password.
	role_id (FK)	integer	No	Role assigned to the account. References roles.
	account_status	enum	No	Whether the account is pending, active, locked or inactive.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.
	last_login	datetime	Yes	Date and time of the last successful sign-in.
	failed_login_attempts	integer	No	Consecutive failed sign-ins; resets on success.
	must_change_password	boolean	No	True when the holder must set a new password before using the system.
	locked_until	datetime	Yes	Account is locked against sign-in until this time.
	device_verification	boolean	No	True when sign-in from a new device needs an emailed code.

Table 2 lists the user roles recognized by the system. Each account is assigned exactly one role, which determines the modules and actions available to that user. This supports role-

based access control across the system.

Table 2.

Roles Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
roles	id (PK)	integer	No	Unique identifier of the record.
	name	enum	No	Role code.
	description	string	Yes	Free-text description.

Table 3 records the active sign-in sessions of users. Only a hashed copy of each session token is stored, together with its expiry, last use, and whether it has been revoked. This allows users to remain signed in securely and allows sessions to be ended when needed.

Table 3.

Refresh Tokens Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
refresh_tokens	id (PK)	integer	No	Unique identifier of the record.
	token	string	No	SHA-256 hash of the refresh token.
	client	string	No	Session type.
	last_used_at	datetime	No	When it was last used.
	user_id (FK)	integer	No	Account the record belongs to. References users.
	expires_at	datetime	No	Date and time this stops being valid.
	created_at	datetime	No	Date and time the record was created.
	revoked	boolean	No	True once the session has been signed out or revoked.

Table 4 keeps a list of browsers and mobile devices that a user has verified through an emailed code. It records the device name, IP address, and validity period of each device. This prevents unrecognized devices from accessing an account without verification.

Table 4.*Trusted Devices Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
trusted_devices	id (PK)	integer	No	Unique identifier of the record.
	user_id (FK)	integer	No	Account the record belongs to. References users.
	token_hash	string	No	SHA-256 hash of the token held by the device.
	label	string	No	Readable device name (browser and operating system).
	ip_address	string	Yes	IP address of the request.
	created_at	datetime	No	Date and time the record was created.
	last_used_at	datetime	No	When it was last used.
	expires_at	datetime	No	Date and time this stops being valid.
	revoked_at	datetime	Yes	When the device was removed from trusted devices.

Table 5 holds pending sign-in verifications sent to a user's email when signing in from a new device. It stores the hashed verification code, the number of attempts, and its expiry. This adds a second layer of protection to the sign-in process.

Table 5.*Login Challenges Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
login_challenges	id (PK)	integer	No	Unique identifier of the record.
	user_id (FK)	integer	No	Account the record belongs to. References users.
	token_hash	string	No	SHA-256 hash of the token held by the device.
	code_hash	string	No	Hash of the 6-digit verification code.
	label	string	No	Readable device name (browser and operating system).
	ip_address	string	Yes	IP address of the request.
	attempts	integer	No	Number of attempts made so far.
	last_sent_at	datetime	No	When the code was last emailed.
	expires_at	datetime	No	Date and time this stops being valid.
	consumed_at	datetime	Yes	When the challenge was completed.
	created_at	datetime	No	Date and time the record was created.

Table 6 records single-use sign-in links that have already been used. Keeping these records prevents the same link from being used more than once.

Table 6.*Used Magic Tokens Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
used_magic_tokens	id (PK)	integer	No	Unique identifier of the record.
	jti	string	No	Unique ID (JWT ID) of the magic-link token.
	user_id	integer	No	Account the record belongs to.
	used_at	datetime	No	When the link was used.
	expires_at	datetime	No	Date and time this stops being valid.

Table 7 records each user's acceptance of the system's Privacy Notice, including the notice version, the date and time of acceptance, and the device used. This serves as evidence of

consent in compliance with the Data Privacy Act of 2012.

Table 7.

Privacy Consents Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
privacy_consent	id (PK)	integer	No	Unique identifier of the record.
	user_id (FK)	integer	No	Account the record belongs to. References users.
	notice_version	string	No	Version of the Privacy Notice that was accepted.
	accepted_at	datetime	No	When the notice was accepted.
	ip_address	string	Yes	IP address of the request.
	user_agent	string	Yes	Browser or app user-agent string.

Table 8 contains requests for new user accounts submitted for approval by the System Administrator. It records the applicant's personal details, intended role, designation, station, and attached Personal Data Sheet, as well as the status of the request. This ensures that account creation follows a controlled approval process.

Table 8.*Account Creation Requests Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
account_creation_requests	id (PK)	integer	No	Unique identifier of the record.
	requested_by_user_id (FK)	integer	No	User (usually AO II) who submitted the request. References users.
	first_name	string	No	Given name.
	last_name	string	No	Surname.
	middle_name	string	Yes	Middle name.
	suffix	string	Yes	Name suffix (Jr., III).
	email	string	No	Email address; unique regardless of letter case.
	birth_date	datetime	Yes	Date of birth.
	gender	enum	Yes	Sex as recorded.
	civil_status	enum	Yes	Civil status.
	contact_number	string	Yes	Mobile or phone number.
	address	string	Yes	Home address.
	role	enum	No	Role the new account will receive.
	designation	string	No	Position / designation title.
	school	string	Yes	School or station of assignment.
	district	string	Yes	School district.
	initial_password	string	No	Temporary password for the first sign-in (must be changed).
	date_hired	datetime	Yes	Date of original appointment.
	pds_original_file_name	string	Yes	Original name of the attached Personal Data Sheet.
	pds_storage_path	string	Yes	Storage path of the attached PDS.
	pds_mime_type	string	Yes	File type of the attached PDS.
	pds_file_size	integer	Yes	Size of the attached PDS in bytes.
	status	enum	No	Current status of the record.
	rejection_reason	string	Yes	Reason given when rejected.
	created_user_id (FK)	integer	Yes	Account that was created when the request was approved. References users.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 9 contains the main profile of every employee, which serves as the basis of the Digital 201 file. It records the employee number, name, personal information, school and district assignment, designation, appointment status, and plantilla position. This provides a single, authoritative record for each employee.

Table 9.

Personnel Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
personnel	id (PK)	integer	No	Unique identifier of the record.
	user_id (FK)	integer	No	Login account of this personnel (one-to-one). References users.
	employee_id	string	No	DepEd employee number; unique.
	first_name	string	No	Given name.
	last_name	string	No	Surname.
	middle_name	string	Yes	Middle name.
	suffix	string	Yes	Name suffix (Jr., III).
	birth_date	datetime	Yes	Date of birth.
	gender	enum	Yes	Sex as recorded.
	civil_status	enum	Yes	Civil status.
	contact_number	string	Yes	Mobile or phone number.
	address	string	Yes	Home address.
	school	string	Yes	School or station of assignment.
	district	string	Yes	School district.
	designation	string	No	Position / designation title.
	date_hired	datetime	Yes	Date of original appointment.
	appointment_status	enum	Yes	Employment status printed on the service record.
	status	enum	No	Current status of the record.
	plantilla_item_id (FK)	integer	Yes	Plantilla item (position) the personnel occupies. References plantilla_items.
	profile_complete	boolean	No	True once the required profile fields are filled in.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 10 stores the documents that make up an employee's Digital 201 file, such as the Personal Data Sheet, certificates, and licenses. It records the document type, file details, issue and expiration dates, review status, and text extracted by OCR. This allows documents to be tracked, reviewed, and replaced without losing earlier versions.

Table 10.*Personnel Files Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
personnel_files	id (PK)	integer	No	Unique identifier of the record.
	personnel_id (FK)	integer	No	Personnel record this row belongs to. References personnel.
	document_type_id	string	No	Code of the 201-file document type (e.g. PDS, SALN).
	document_type_name	string	No	Display name of the document type.
	original_file_name	string	Yes	File name as uploaded.
	stored_file_name	string	Yes	File name in storage.
	storage_path	string	Yes	Path of the file in storage.
	mime_type	string	Yes	File type (MIME).
	file_size	integer	Yes	File size in bytes.
	is_required	boolean	No	True if this document is required in the 201 file.
	issue_date	date	Yes	Date the document was issued.
	expiration_date	date	Yes	Date the document expires (licences, certificates).
	remarks	string	Yes	Free-text remarks.
	status	enum	No	Current status of the record.
	rejection_reason	string	Yes	Reason given when rejected.
	reviewed_at	datetime	Yes	When the document was reviewed.
	reviewed_by_user_id (FK)	integer	Yes	User who reviewed the document. References users.
	replaces_document_id (FK)	integer	Yes	Earlier document this upload supersedes. References personnel_files.
	ocr_extracted_data_json	json	Yes	Fields extracted by OCR.
	ocr_confidence_score	decimal	Yes	OCR confidence from 0 to 1.
	ocr_status	string	Yes	State of OCR processing.
	ocr_processed_at	datetime	Yes	When OCR finished.
	ocr_error_message	string	Yes	OCR error, if any.
	deleted_at	datetime	Yes	Soft-delete time; empty while the record is active.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 11 lists the authorized positions of the Schools Division. Each item records its item number, position title, salary grade, department, and division. An item is considered occupied when it is assigned to an employee.

Table 11.

Plantilla Items Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
plantilla_items	id (PK)	integer	No	Unique identifier of the record.
	item_number	string	No	Official plantilla item number; unique.
	position_title	string	No	Title of the position.
	salary_grade	integer	No	Salary grade of the position.
	department	string	No	Department / office.
	division	string	No	Schools division.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 12 records the career events of each employee, such as promotions, trainings, awards, and changes of designation. Each entry may be linked to a supporting document. This provides a chronological record of an employee's service.

Table 12.*Career History Entries Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
career_history_entries	id (PK)	integer	No	Unique identifier of the record.
	personnel_id (FK)	integer	No	Personnel record this row belongs to. References personnel.
	event_type	enum	No	Kind of career event.
	event_date	datetime	No	Date of the event.
	details_json	json	Yes	Additional details (JSON).
	document_id (FK)	integer	Yes	Related uploaded document. References uploaded_documents.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 13 defines each promotion or reclassification cycle, including its type, schedule, status, and the ranking criteria used. This allows the HRMO to manage promotion periods in an organized manner.

Table 13.*Promotion Cycles Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
promotion_cycles	id (PK)	integer	No	Unique identifier of the record.
	name	string	No	Name of the item.
	type	enum	No	Type / category of the record.
	start_date	datetime	No	Start of the cycle.
	end_date	datetime	No	End of the cycle.
	status	enum	No	Current status of the record.
	rules_configuration_json	json	Yes	Ranking criteria and weights for the cycle.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 14 records the applications of employees to a promotion cycle, including the applicant number, status, scores per criterion, and final rank. This supports a transparent comparative assessment of applicants.

Table 14.

Promotion Applications Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
promotion_applications	id (PK)	integer	No	Unique identifier of the record.
	applicant_number	string	Yes	Applicant number assigned for the cycle; unique.
	personnel_id (FK)	integer	No	Personnel record this row belongs to. References personnel.
	promotion_cycle_id (FK)	integer	No	Promotion cycle applied to. References promotion_cycles.
	status	enum	No	Current status of the record.
	application_date	datetime	No	When the application was filed.
	final_rank	integer	Yes	Final rank after comparative assessment.
	score_details_json	json	Yes	Breakdown of scores per criterion.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 15 records each HR transaction filed by or for an employee as it moves from submission, to validation by the Administrative Officer II, to approval by the HRMO. It tracks the transaction type, current status, assigned officer, deadlines, and the number of resubmissions. This ensures that every request is monitored until it is completed.

Table 15.*Transactions Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
transactions	id (PK)	integer	No	Unique identifier of the record.
	personnel_id (FK)	integer	No	Personnel record this row belongs to. References personnel.
	transaction_type_id (FK)	integer	No	Kind of HR transaction. References transaction_types.
	status	enum	No	Workflow stage of the transaction.
	submission_date	datetime	Yes	When submitted for validation.
	validation_date	datetime	Yes	When validated.
	approval_date	datetime	Yes	When approved by HRMO.
	remarks	string	Yes	Free-text remarks.
	current_assignee_id (FK)	integer	Yes	User whose queue the transaction is in. References users.
	resubmission_count	integer	No	Number of times returned and resubmitted.
	stage_entered_at	datetime	Yes	When the transaction entered its current status.
	escalated_at	datetime	Yes	When repeated corrections escalated it to HRMO.
	escalation_reviewed_at	datetime	Yes	When HRMO returned the escalated transaction with instructions.
	deficiency_deadline	datetime	Yes	Deadline for correcting a deficiency.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 16 lists the kinds of HR transactions that can be filed in the system. Each type determines the set of documentary requirements needed for that transaction.

Table 16.*Transaction Types Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
transaction_types	id (PK)	integer	No	Unique identifier of the record.
	name	string	No	Name of the item.
	description	string	Yes	Free-text description.

Table 17 stores the files submitted to satisfy the requirements of a transaction. It records the file details, the uploader, the OCR results, and the validation result and notes. This allows each requirement to be checked and validated individually.

Table 17.*Uploaded Documents Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
uploaded_documents	id (PK)	integer	No	Unique identifier of the record.
	transaction_id (FK)	integer	No	Transaction this row belongs to. References transactions.
	requirement_template_id (FK)	integer	No	Requirement this file satisfies. References requirement_templates.
	storage_path	string	No	Path of the file in storage.
	file_name	string	No	Name of the uploaded file.
	file_size	integer	Yes	File size in bytes.
	mime_type	string	Yes	File type (MIME).
	file_hash	string	Yes	SHA-256 hash of the file, used to detect duplicates.
	uploaded_by_user_id (FK)	integer	No	User who uploaded the file. References users.
	upload_date	datetime	No	When the file was uploaded.
	status	enum	No	Current status of the record.
	ocr_extracted_data_json	json	Yes	Fields extracted by OCR.
	ocr_confidence_score	decimal	Yes	OCR confidence from 0 to 1.
	corrected_ocr_data_json	json	Yes	OCR fields after manual correction.
	validation_notes	string	Yes	Notes from the validator.
	validated_by_user_id (FK)	integer	Yes	User who validated the document. References users.
	validation_date	datetime	Yes	When validated.
	is_duplicate	boolean	No	True if the same file was already uploaded.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 18 defines the documentary requirements of each transaction type, including whether each requirement is mandatory and which fields should be extracted through OCR. This standardizes what employees need to submit.

Table 18.*Requirement Templates Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
requirement_templates	id (PK)	integer	No	Unique identifier of the record.
	transaction_type_id (FK)	integer	No	Kind of HR transaction. References transaction_types.
	name	string	No	Name of the item.
	description	string	Yes	Free-text description.
	is_mandatory	boolean	No	True if the requirement must be submitted.
	expected_data_type	string	No	Kind of document expected (e.g. PDF, image).
	ocr_fields_json	json	Yes	Fields OCR should extract from this requirement.
	created_at	datetime	No	Date and time the record was created.
	updated_at	datetime	No	Date and time the record was last updated.

Table 19 keeps a snapshot of an uploaded document each time it changes. This preserves the history of a document for reference and auditing.

Table 19.*Document Revisions Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
document_revisions	id (PK)	string	No	Unique identifier of the record.
	document_id (FK)	integer	No	Related uploaded document. References uploaded_documents.
	snapshot	json	No	Full copy of the document record at that point.
	created_at	datetime	No	Date and time the record was created.

Table 20 saves unfinished forms, such as the Personal Data Sheet, while a transaction is being prepared. This allows users to continue their work later without losing what they have

entered.

Table 20.

Form Drafts Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
form_drafts	transaction_id (PK, FK)	integer	No	Transaction this row belongs to. References transactions.
	template_id (PK)	string	No	Form template being drafted (e.g. PDS).
	version	string	No	Version of the form template.
	payload	json	No	Saved data (JSON).
	updated_at	datetime	No	Date and time the record was last updated.

Table 21 serves as the audit trail of the system. It records who performed an action, what was done, on which record, when, from where, and with what outcome. Each entry is linked to the previous one through a hash, which makes tampering detectable.

Table 21.*Validation Logs Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
validation_logs	id (PK)	integer	No	Unique identifier of the record.
	entity_type	string	No	Kind of record acted on.
	entity_id	integer	No	ID of the record acted on.
	action	string	No	Action code (e.g. LOGIN, TRANSACTION_APPROVED).
	details_json	json	Yes	Additional details (JSON).
	user_id (FK)	integer	No	Account the record belongs to. References users.
	ip_address	string	Yes	IP address of the request.
	user_agent	string	Yes	Browser or app user-agent string.
	status	string	No	Current status of the record.
	timestamp	datetime	No	When the action happened.
	category	string	Yes	Audit category (Authentication, Documents, ...).
	severity	string	Yes	INFO, WARNING or CRITICAL.
	outcome	string	Yes	SUCCESS, FAILURE or DENIED.
	action_label	string	Yes	Readable action name.
	target_type	string	Yes	Kind of target record.
	target_id	integer	Yes	ID of the target record.
	target_reference	string	Yes	Readable reference of the target.
	actor_email	string	Yes	Email of the actor at the time.
	actor_role	string	Yes	Role of the actor at the time.
	actor_station	string	Yes	Station of the actor at the time.
	client_source	string	Yes	web or mobile.
	request_id	string	Yes	ID tying entries from one request together.
	failure_reason	string	Yes	Why the action failed.
	record_hash	string	Yes	Hash of this audit entry.
	previous_hash	string	Yes	Hash of the previous entry (chain link).
	hash_version	integer	Yes	Version of the hashing scheme.

Table 22 stores the in-app notifications sent to users, such as updates on the status of their transactions. It records the message, its type, and whether it has been read.

Table 22.*Notifications Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
notifications	id (PK)	integer	No	Unique identifier of the record.
	user_id (FK)	integer	No	Account the record belongs to. References users.
	message	string	No	Notification text.
	type	enum	No	Type / category of the record.
	is_read	boolean	No	True once the user has opened the notification.
	created_at	datetime	No	Date and time the record was created.
	related_entity_id	integer	Yes	ID of the record the notification is about.
	related_entity_type	string	Yes	Kind of record the notification is about.

Table 23 holds pending background tasks, such as email notifications, produced by workflow actions. Each task is processed only once and is retried if it fails, which ensures that notifications are delivered reliably.

Table 23.*Workflow Outbox Data Dictionary*

Table Name	Field Name	Data Type	Allow Nulls	Field Description
workflow_outbox	id (PK)	string	No	Unique identifier of the record.
	event_key	string	No	Unique key so each event is processed only once.
	kind	string	No	Kind of event or run.
	payload	json	No	Saved data (JSON).
	attempts	integer	No	Number of attempts made so far.
	available_at	datetime	No	Earliest time the event may be processed.
	processed_at	datetime	Yes	When processing finished.
	last_error	string	Yes	Last processing error.
	created_at	datetime	No	Date and time the record was created.

Table 24 records the results of each database backup and restore test, including the time it ran, its size, whether it was encrypted and verified, and any errors. This provides evidence that the system's data is backed up and recoverable.

Table 24.

Backup Runs Data Dictionary

Table Name	Field Name	Data Type	Allow Nulls	Field Description
backup_runs	id (PK)	integer	No	Unique identifier of the record.
	run_key	string	No	Unique key of the backup run (makes re-sent reports idempotent).
	kind	string	No	Kind of event or run.
	status	string	No	Current status of the record.
	started_at	datetime	Yes	When the run started.
	finished_at	datetime	Yes	When the run finished.
	size_bytes	bigint	Yes	Backup size in bytes.
	encrypted	boolean	Yes	True if the backup is encrypted.
	manifest_verified	boolean	Yes	True if the backup manifest was verified.
	missing_objects	integer	Yes	Storage objects missing from the backup.
	object_count	integer	Yes	Storage objects in the backup.
	retention_days	integer	Yes	Days the backup is kept.
	error	string	Yes	Error message if the run failed.
	source	string	Yes	Job that reported the run.
	reported_at	datetime	No	When the result was reported.